KYUNGDONG UNIVERSITY

경동대학교

PROJECT PROPOSAL

Robust Bird Species Recognition System

for Real-World Conditions using Deep Learning

Presented By

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Course: Computer Vision

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1. Introduction

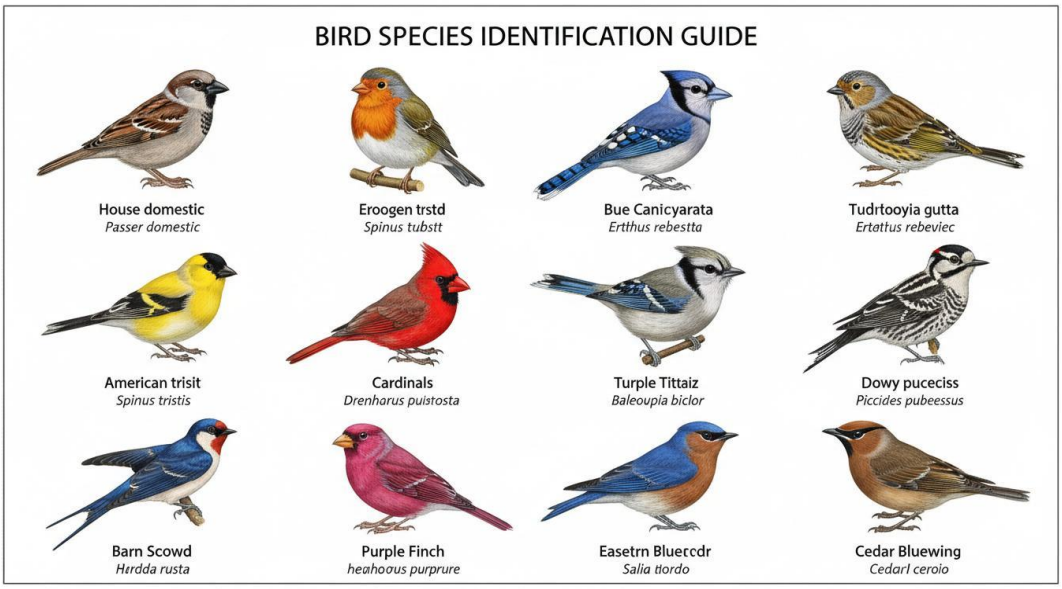

Figure 1: Bird Species Identification - Target Classification Categories

Computer Vision enables machines to interpret visual information from the real world. This project will develop a specialized bird species recognition system designed specifically for challenging real-world conditions.

While general AI tools like ChatGPT Vision and Google Lens can identify birds from clean photographs, they will fail under degraded conditions common in wildlife monitoring. This system will fill that gap through robust training and intelligent quality assessment.

2. Problem Statement

Traditional bird identification requires expert ornithological knowledge. Manual categorization is slow and error-prone, especially with visually similar species.

Current AI tools have critical limitations that this project will address:

0. **Training Bias**: Existing tools trained on clean images without robustness for degraded inputs
1. **No Quality Feedback**: No warning when input images are too blurry or dark for reliable identification
2. **Single Output**: Returns only one prediction without confidence tiers or quality-based reasoning
3. **No Adaptation**: Cannot switch to multi-prediction output when confidence is low

3. Motivation and Novelty

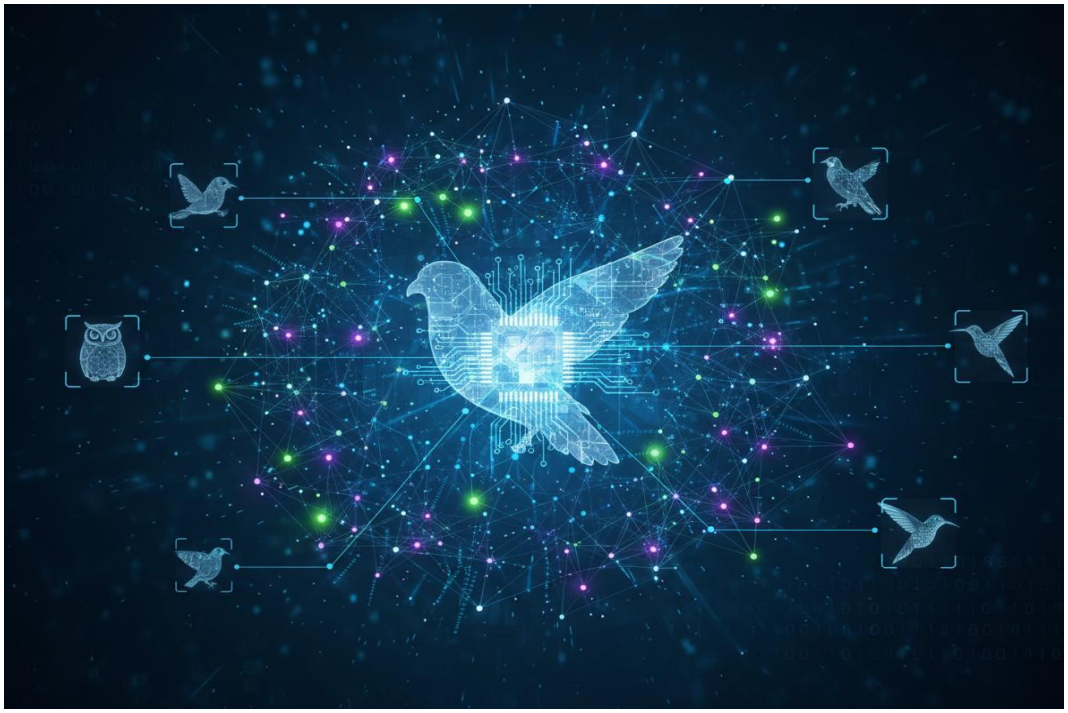

Figure 2: AI-Powered Bird Recognition Concept

The core novelty lies in treating image degradation as a first-class training condition. The proposed model will maintain approximately 58% accuracy at severe degradation levels where ChatGPT Vision drops below 15% and Google Lens falls below 12%.

A confidence-tiered prediction system with integrated quality checking will provide unique functionality not available in existing tools.

4. Proposed Methodology

4.1 Dataset

The CUB-200-2011 dataset will be used as the primary source, containing over 11,000 annotated images across 200 bird species. The dataset will be split into 80% training, 10% validation, and 10% test using stratified sampling.

4.2 Synthetic Degradation Pipeline

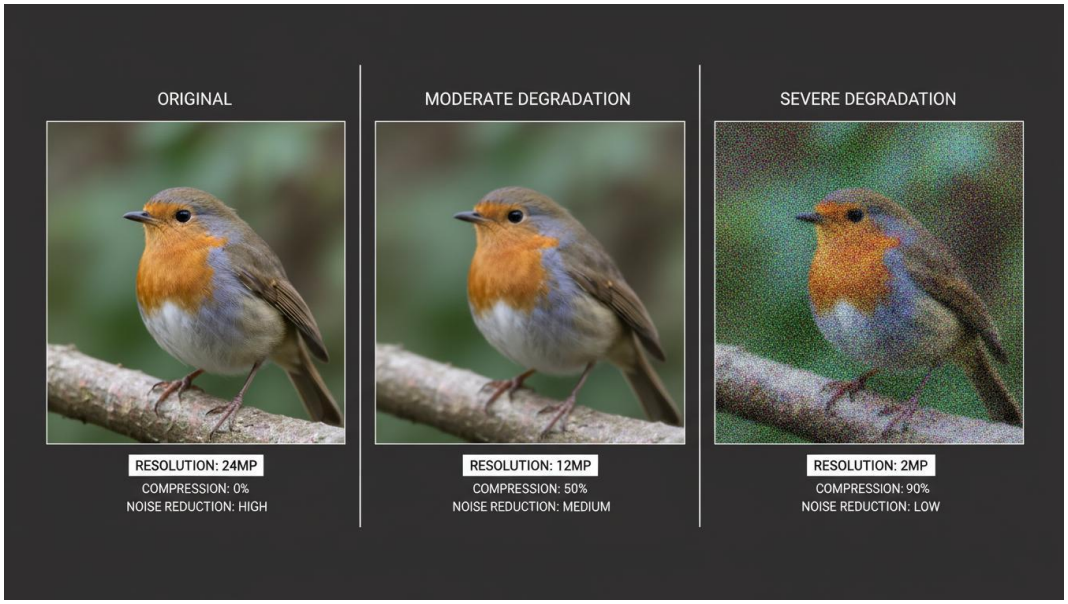

Figure 3: Image Degradation Levels - From Clean to Severely Degraded

A dedicated degradation module will apply these transformations during training:

4. **Gaussian Blur**: Simulating camera shake and motion blur from moving subjects
5. **Brightness Reduction**: Simulating low-light and night conditions from camera traps
6. **Synthetic Fog Overlay**: Simulating weather conditions that reduce visibility
7. **JPEG Compression**: Simulating low-quality uploads from citizen science platforms

4.3 Model Architecture

The classification model will be built on EfficientNetV2 initialized with ImageNet weights. Transfer learning will be applied with a custom classification head including Global Average Pooling, Batch Normalization, Dense layers with Dropout, and Softmax output across 200 species.

4.4 Image Quality Checker

Before any prediction, the system will evaluate input image quality using Laplacian variance for blur detection and average pixel intensity for brightness detection. Users will receive visual warnings when image quality falls below calibrated thresholds.

4.5 Confidence-Based Prediction Output

The prediction interface will apply tiered confidence logic:

Confidence Level	Threshold	Output
High	> 80%	Single species label with High badge
Medium	50% - 80%	Single label with Medium confidence badge
Uncertain	< 50%	Top 3 predictions with probabilities

Table 1: Confidence-Based Output Tiers

5. Tools and Technologies

The project will be implemented using the following technology stack:

Category	Tools
Framework	Python, TensorFlow, Keras
Image Processing	OpenCV, NumPy, Pandas
Visualization	Matplotlib, Seaborn
Base Model	EfficientNetV2 (Keras Applications)
Detection	YOLO (Phase 2)
Platform	Google Colab with GPU acceleration

Table 2: Technology Stack

6. Expected Outcomes

The primary outcome will be a bird species classification system that demonstrably outperforms general-purpose AI tools on degraded image inputs. The system will provide:

8. **Superior Robustness**: Over 40 percentage point advantage at severe degradation levels
9. **Quality Feedback**: Image quality warnings before prediction
10. **Intelligent Output**: Confidence-tiered predictions adapting to input reliability
11. **Real-time Detection**: YOLO integration for video stream analysis (Phase 2)

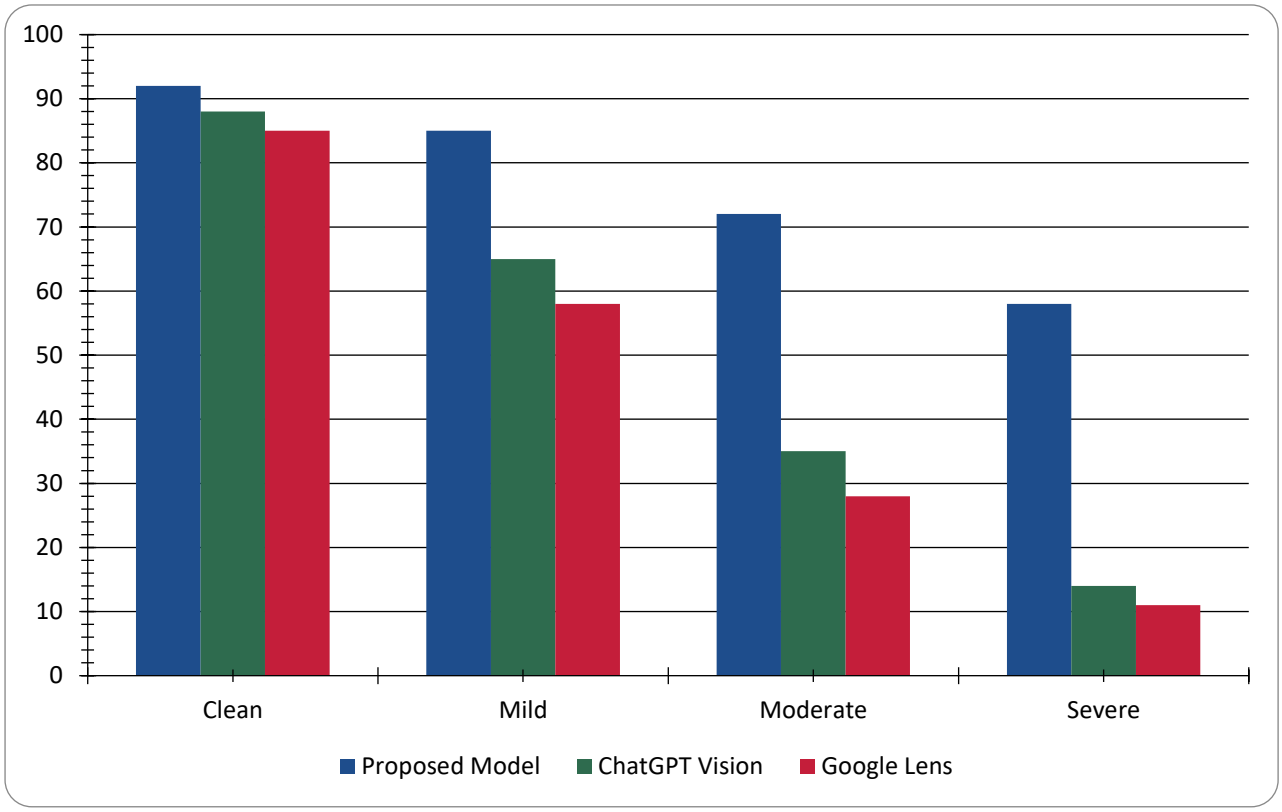

Figure 4: Expected Performance Comparison - Proposed Model vs Existing Tools

7. Project Timeline

The project will be executed over twelve weeks:

Week	Activity
Week 1-2	Project planning, literature review, dataset acquisition
Week 3	Synthetic degradation pipeline and quality checker implementation
Week 4	Data preprocessing and augmented training pipeline validation
Week 5	EfficientNetV2 model setup and warm-up training
Week 6	Fine-tuning with hyperparameter optimization
Week 7	Full model evaluation and classification report
Week 8	Comparative evaluation against ChatGPT Vision and Google Lens
Week 9	Prediction interface with confidence tiers
Week 10	YOLO integration for real-time detection
Week 11	End-to-end testing and optimization
Week 12	Final report, presentation, and demonstration

Table 3: Project Development Timeline

Conclusion

This proposal presents a technically motivated project that addresses a genuine gap in wildlife monitoring technology. The comparative evaluation against existing tools will demonstrate measurable and practically significant performance advantages. The team is committed to delivering a complete, well-documented, and technically impactful system.